



University of East London

**Assignment Title: Mental Wealth: Professional Life
(Data Ecology) Evaluative Review**

Course: DS7001 2526 (T1) Mental Wealth: Professional life

Instructor: Dr. Yang Li

Student Name: Keshav Poudel

Student ID: 3064995

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Part 1: How the UK Government Should Promote and Regulate AI-Driven Mental Health Tools Used by the NHS: Balancing Innovation, Safety and Social Wellbeing?

Abstract

The UK is dealing with more and more mental health problems, such as rising levels of anxiety, depression, and loneliness. At the same time, the NHS is still having trouble meeting the needs of all its patients. As Consequently, artificial intelligence has been increasingly used by NHS mental health services to support early intervention, symptom monitoring, and triage.

These technologies could be helpful in some ways, like making it easier for people to get care, cutting down on wait times, and providing support that can grow with the needs of the patients. However, they also raise serious concerns about patient safety, data privacy, bias, and relying too much on automated systems.

How the UK government should encourage and regulate AI-powered mental health solutions used by the NHS is examined in this evaluative evaluation. The study critically assesses the advantages and disadvantages of AI-enabled mental health support using scholarly literature, policy studies, regulatory guidelines, and recent media investigations. Sensitive emotional data is gathered, processed, and governed using a data ecology method. The impact of these systems on social behaviour and public trust is also examined.

According to the report, government policy must strike a balance between novel concepts and robust safeguards. Evidence-based strategies that increase access and early intervention should be the main emphasis of effective promotion. On the other hand, regulations should guarantee accountability, data security, and transparent human monitoring. The study recommends directions for further research on trust, equity, and long-term societal implications and says that AI should be a helpful supplement to human care rather than a replacement for it.

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1. Introduction

More and more people in the UK are having mental health problems. The Mental Health Foundation says that a lot of people in society still suffer from anxiety, depression, and stress, and loneliness has spread to people of all ages (Mental Health UK, 2025). The Office for National Statistics (2023) said that 7% of adults in the UK are chronically lonely, with young adults and people who live in cities being the most affected (Office for National Statistics, 2023). People who feel lonely are more likely to get depressed and hurt themselves or use unhealthy ways to deal with their feelings. These new trends are making things harder for the NHS, which is already having trouble keeping up with demand for therapy and mental health staff. The NHS is also having to deal with longer wait times for therapy and more need for early intervention services.

Simple rule-based chatbots that adhered to pre-written dialogue scripts were utilized in the creation of the earliest computerized cognitive behavioural therapy (CBT) applications. Despite their limitations, the existing tools demonstrated that structured psychological support could nevertheless be provided online. Natural language processing (NLP) and machine learning systems have advanced significantly in the last ten years. AI-based platforms Wysa and Woebot assist mental health services by analysing user feedback. The NHS has approved Limbic Access as a triage system that does the same thing(woebothealth, 2024) (Limbic , 2024). AI and mental health technologies are crucial for managing NHS services since they aid with the growing number of patients, facilitate patient communication with professionals, and improve the initial evaluation procedures, according to a 2025 POSTnote from the UK Parliament(Hannah Gardiner and Natasha Mutebi , 2025).

The primary organization involved in the creation and use of digital mental health technologies is the NHS. The NHS Talking Therapies initiative assists healthcare organizations in utilizing digital platforms that facilitate mental health care, reduce patient wait times, and facilitate physician assessments and referrals(NHS England, 2024). In modern healthcare, digital tools are mostly used for three things: assessments, remote therapy sessions, and keeping track of ongoing symptoms. Local NHS trusts have added AI technology to help their staff figure out which referrals need to be dealt with right away while they look for signs of danger and deal with longer patient wait times.

The research examines AI-driven mental health solutions employed by the NHS, as three critical factors have been identified from both academic research and practical implementations. The primary benefit of mental health services utilizing AI technology is its capacity to ensure equitable access for individuals encountering difficulties in obtaining these services, as AI offers immediate, non-judgmental support via digital platforms that function independently of traditional healthcare institutions(Mental Health UK, 2025). People usually start with digital tools because they are both private and easy to get to. Second, the NHS's main way of working is on public trust. AI tools necessitate users to adhere to fundamental ethical responsibilities due to their operation with individuals possessing limited safeguards. Third research findings indicate that AI technology poses significant threats to mental health services. According to Mental Health UK (2025), AI chatbots give users medical advice that hasn't been checked. The Guardian (2025) and Reuters (2025) both did research that showed that general-purpose AI systems make harmful and inappropriate content when users share their upsetting

experiences (Booth, 2025) (Rozen, 2025). These results demonstrate the necessity of governmental oversight.

AI has long been used in healthcare. In the 1970s, the MYCIN diagnostic model demonstrated how computers could assist physicians in making medical choices. As machine learning technology advanced in the 1990s and 2000s, new diagnostic systems were developed for use by pathologists, geneticists, and radiologists. Because developers had to learn about people's emotions and how to handle complex emotional and linguistic data, creating mental health apps took longer. These days, AI mental health applications range from simple symptom assessment tools to more sophisticated conversational systems that behave like friends. Despite their increased operational capabilities, experts remain uncertain about their strategic stance. In the medical industry, AI systems ought to assist physicians in making judgments rather than making them themselves. The healthcare system needs to make AI technology more available and take strong steps to keep patients safe and make sure that these systems are used responsibly (Royal College of Psychiatrists, 2024).

The widespread usage of AI-based mental health solutions presents significant concerns around system safety, equitable treatment, privacy protection, and public monitoring. Since the NHS is the biggest public health agency in the UK, it must closely monitor these issues. The government determines which instruments to employ by establishing regulations and ensuring that the public continues to have faith in them.

The UK government should support AI tools that increase accessibility to mental health care, but it must also establish regulations that protect patients and maintain public confidence. The report substantiates its argument by examining academic research, NHS digital service policy, and recent advancements in the system. The assessment examines the advantages and disadvantages of AI mental health tools, their impact on data systems, and provides guidelines for the ethical use of AI to ensure the safety of individuals and society.

2. Methodology

The research utilizes critical literature review methodologies to analyse AI applications in NHS mental health treatment and determine appropriate measures for the UK Government. The objective extends beyond merely delineating AI technologies. It also entails examining their positive and negative aspects and their potential impact on society at large. The review achieves its objective by examining academic research on digital mental health interventions, official governmental documents, UK legislative records, guidance from mental health organizations, and news articles concerning emerging issues related to AI behaviour.

The UK Parliament POSTnote regarding AI applications in mental healthcare, in conjunction with reports from the Mental Health Foundation, Mental Health UK, and NHS digital treatment tool guidelines, is a collection of significant materials. The research substantiates its assertions with peer-reviewed studies examining the impact of conversational AI on mental health, as well as the efficacy of AI therapy support and early intervention technologies. To identify current threats, they examine news investigations and regulatory updates. These dangers encompass hazardous chatbot responses and data security violations.

The evidence for AI-driven mental health support systems is evaluated concerning research that underscores safety issues, alongside ethical and technical hurdles. The paper examines the dynamics of data flow inside AI systems, their decision-making processes, and the implications of these decisions on individuals. The research approach enables the analysis of AI's influence on mental health treatment systems that extend beyond basic medical outcomes.

This review-based methodology is appropriate for evaluating AI in NHS mental health care because of the technology's swift advancement and its incorporation within complex social, ethical, and institutional contexts. This method facilitates a comprehensive assessment of both benefits and drawbacks by synthesizing evidence from academic research, governmental papers, and real-world instances, rather than depending exclusively on technical analysis. However, the methodology is limited by its reliance on secondary sources and the absence of primary empirical data from patients or professionals. Thus, the conclusions reflect pre-existing facts and documented results rather than direct evaluations of user experiences. Despite these issues, the strategy aligns with the objectives of the Mental Wealth module as it emphasizes responsible data utilization and its societal implications.

3. Key Components of AI in NHS Mental Health Services: Advantages, Dangers, and Government Responsibilities

3.1. Government Assistance for Artificial Intelligence in NHS Mental Health

The UK Government has selected artificial intelligence as the primary method to transform the operations of the NHS. The AI in Health and Care Award facilitated the development of Limbic Access, an AI tool designed to assist in mental health treatment triage. The NHS Long Term Plan asserts that digital innovation is crucial for enhancing mental health services for patients.

It is hard to deliver mental health services because patients have to wait a long time for treatment while the healthcare system tries to keep enough staff on hand. AI systems do three main things: they do first evaluations, recommend services, and screen referrals to help people find the best care options. The system makes it easier for medical staff to do their jobs and speeds up the delivery of services. AI chatbots and self-help tools work together to provide support services to people who need help at night or who want to avoid face-to-face help.

AI tools help people figure out if they need to see a doctor for their symptoms by starting their health assessment journey. An AI tool helps users figure out what they need by asking them organized questions. This helps them decide when they need to get in touch with a professional. The Mental Health Foundation says that kids need help right away because it stops their problems from getting worse in the future.

Advantages of Government Endorsement

- The system must provide mental health services for individuals requiring alternatives to conventional office-based care.
- Automated systems manage the administrative responsibilities of NHS personnel, thereby facilitating their work.
- There are more help services available now because they are open when regular services are not.
- The system's fast triage system helps users get immediate service by directing them to the right care options.
- The program helps people with mental health problems who live in rural areas where mental health services are hard to get to.

The government support system makes it easier for people to get medical care, strengthens initial treatment programs, and reduces the need for medical staff to work.

Risks of Government Promotion Without Safeguards

The government's promotional activities create possible risks that need good protection systems to work well. These risks are:

- The system relies too much on automated tools, which means that patients don't interact with people as much.
- The quality of AI systems is still not the same for all AI systems.
- This situation will make the digital divide between people who can get online and those who can't even bigger.
- People trust AI too much because they don't understand it well enough.

Because uncontrolled promotion could lead to unsafe and unfair practices, promotion strategies need to be carefully watched. The evidence shows that problems with NHS staffing and funding are still not being fixed, even though automation is becoming more common. AI systems are helpful tools that make human care better, but they can't replace long-term solutions that increase the number of workers and the availability of healthcare for people.

3.2. Government Regulation and Oversight in NHS Mental Health AI

As AI is used more and more to help people with mental health issues, rules need to change too. The UK Government, the NHS, the Information Commissioner's Office, and developers all have a part to play in making sure that AI tools don't hurt people.

Data Safeguarding and Confidentiality

AI mental health tools collect highly confidential information about individuals. Discussions may encompass trauma, interpersonal relationships, self-injury, substance abuse, or intrusive thoughts. Effective supervision prevents AI systems from supplanting critical human discernment. The Information Commissioner's Office emphasizes the significance of explicit consent, secure storage, and data minimization. NHS-approved digital products must adhere to stringent requirements. Nonetheless, private platforms may not consistently adhere to equivalent requirements.

Many individuals are concerned about privacy. Mental Health UK says that a lot of people who use the service are unsure about how their data is stored and whether it could be used for advertising or sold to other companies. Trust in digital mental health tools can quickly go down if there isn't clear transparency. The government must implement stringent regulations to prevent the misuse of sensitive information.

Clear communication and openness

People need to know when they are talking to an AI system instead of a real person. Some AI tools can give answers that seem very personal or understanding, which can make people think they are talking to a therapist. To prevent misunderstandings and manage user expectations regarding the system's capabilities, transparency is essential. Government legislation and NHS guidelines stipulate that digital mental health solutions must explicitly delineate:

- the capabilities and limitations of the AI
- the circumstances necessitating professional assistance
- the utilization of data
- the dangers that users should acknowledge

If the tool's purpose is not clearly defined, users may misinterpret its function and utilize it incorrectly.

Safety and Quality Assurance in the Clinic

The efficacy and safety of AI in mental health must be rigorously evaluated. Mental health exams differ from physical health evaluations as they consider individuals' emotions and thoughts. AI systems may misinterpret content or fail to detect subtle indicators of distress.

Reuters discovered that certain chatbots designed for general use provided harmful or perilous advice to users who expressed intentions of self-harm (Booth, 2025). These results show why the government needs to require strict clinical testing.

The NHS already has rules about what medical devices and digital health tools must do. However, AI systems that evolve over time may require continuous monitoring. Autonomous data-gathering tools may not function as anticipated. So, the government needs to set up systems that make sure audits and performance reviews happen on a regular basis.

Human Oversight and Pathways for Referrals

AI cannot take the place of trained doctors. If AI systems are used to figure out what people need or give them initial advice, they need to have a clear way for people to get help from a person. The NHS asserts that digital tools ought to complement, rather than supplant, face-to-face care. Human monitoring ensures that:

- Crisis situations are identified and addressed promptly
- Users do not become overly accustomed to automated interactions
- Individuals with complex mental health requirements receive appropriate care.

Advantages of Regulation:

- It safeguards individuals from misleading or detrimental material
- It enhances the likelihood of individuals trusting digital mental health resources
- It promotes responsible development among private enterprises.
- It ensures equitable treatment for all demographic groupings.

Disadvantages of Regulation:

- Compliance with regulations may incur significant costs for smaller developers.
- Excessively stringent regulations may impede innovation.
- Rapidly evolving technology may struggle to comply with regulations.

The challenge lies in establishing regulations that adequately ensure user safety while fostering innovation.

3.3. Dangers of Inadequate or Insufficient Regulation

Inadequate governmental oversight of AI mental health tools significantly increases associated hazards.

Excessive Reliance on Artificial Intelligence: In lieu of Human Care Some individuals may exclusively communicate with AI and refrain from seeking assistance from a human being. This is particularly alarming for those with complex diseases. An AI tool may appear empathetic, although it cannot substitute for a therapist.

Inaccurate or Misguided Counsel: AI systems may occasionally provide advice that is neither beneficial nor safe. The risks increase if they are not adequately tested. Research indicates that AI systems may misinterpret emotional cues or provide erroneous guidance in adverse situations.

Violation of Privacy: Inadequate regulation increases the likelihood of personal data being disclosed or compromised. The confidentiality of mental health data renders breaches particularly detrimental to individuals.

Digital Disparity: If AI tools become the primary method via which the NHS assists individuals with mental health issues, those lacking internet access or computer proficiency may be excluded. This may exacerbate the existing gaps.

Public Distrust: Individuals may refrain from utilizing digital services if they lack confidence in AI mental health solutions, despite the potential benefits of such services.

3.4. Beneficial Effects of Robust Promotion and Regulation

The advantages of employing AI in NHS mental health services significantly increase with governmental cooperation and robust regulatory safeguards. Users can get safe and reliable early help, and digital tools can help people find the right NHS services more quickly. This might help people who are lonely or who have to wait a long time to get care stop doing things that are bad for them. Clinicians can spend less time on administrative triage tasks and more time providing complex, patient-centred care at the organizational level. Strong oversight also builds trust in the public, which encourages people to use digital mental health tools responsibly. Equitable governance ensures that AI enhances human care rather than supplanting it.

3.4.1. Recommended Strategic Policy Actions for the Government

The UK Government should implement a series of targeted and aggressive legislative steps to ensure that AI enhances NHS mental health services rather than detracts from them. The government should initially establish AI safety sandboxes for mental health applications. This would permit the evaluation of novel tools in regulated NHS environments prior to widespread implementation. These sandboxes would enable physicians, regulators, and users to identify problems associated with accuracy, bias, and crisis management without jeopardizing vulnerable populations.

Secondly, any NHS-sanctioned AI mental health solutions must incorporate established thresholds for when an individual requires human interaction. These thresholds indicate that AI should serve solely as an adjunct to clinical care rather than as a substitute. Users exhibiting indications of significant distress, self-harm, or emotional dependency must be promptly directed to a qualified professional.

The government should invest in public AI literacy initiatives linked to NHS digital services. Users could make more informed decisions if they were aware of the precise capabilities and limitations of AI systems. This would enhance individuals' trust in digital mental health care and reduce dependence on automated guidance.

NHS procurement contracts must ensure adherence to ethical data management guidelines. These regulations should prohibit the commercial reutilization of emotional data, ensure explicit consent for model training, and clarify data storage and sharing protocols. These measures would safeguard patient health and public trust while facilitating the emergence of innovative concepts.

4. Data Ecology and Societal Impacts

To comprehensively comprehend AI in NHS mental health treatments, one must examine factors outside its efficacy. It requires an examination of data flow, control mechanisms, meaning assignment, and the system's influence on individuals and society. This perspective aligns with data ecology, which considers data systems as essential elements of wider social, ethical, and political frameworks. The integration of AI in mental health involves sensitive data, vulnerable individuals, and powerful organizations, thus necessitating a comprehensive ecological investigation.

4.1. Data Acquisition and Origins

AI mental health tools gather diverse data kinds. These comprise users' written narratives regarding their feelings, behavioural habits, usage frequency, symptom trends, and occasionally demographic information. In NHS environments, AI may have access to referral forms, medical histories, or clinical notes, contingent upon the degree of integration achieved. AI systems can generate personalized replies and identify patterns due to the richness of various data sources.

Individuals seeking emotional help frequently disclose highly confidential information. Individuals may refrain from disclosing such matters to their family, employer, or physician. Certain individuals see a diminished capacity for judgment while utilizing AI tools, potentially resulting in them articulating more than intended. In the absence of effective governance, the collection of private information may result in power disparities. Individuals requiring assistance may lack knowledge on the utilization of their data and the entities with access to it.

The quality of the data significantly influences the system's performance. An AI trained on incomplete, biased, or unrepresentative data may fail to produce reliable interpretations. Mental Health UK asserts that AI techniques may fail to recognize distress in individuals from cultural or linguistic groups that are underrepresented in training datasets. This may result in guidance that is impolite or unhelpful. From a data ecology perspective, quality transcends mere technicality; it is fundamentally a social concern. The system's comprehension of users is influenced by decisions regarding the data to be collected, the sources of that data, and the frequency of collection.

4.2. Data Processing and Model Functionality

Subsequent to collection, data must be processed, analysed, and transformed into model outputs. In traditional clinical environments, human practitioners interpret information via training, empathy, and ethical judgment. AI systems use algorithms to find patterns in language and behaviour. They don't get emotions or context the way people do. AI uses statistical relationships to guess what people are likely to say or what risks they might face.

This makes people worried in places where emotions are important. For instance, people might misunderstand when someone is being sarcastic or sad. Some AI tools might not pick up on small signs that doctors do. Some people may overreact to things that aren't dangerous. Users may not like the fact that the responses are different each time, and it could be dangerous if a system doesn't pick up on distress at the crisis level.

AI systems also make outputs based on the data they were trained on. If the data is biased, those biases get into the system. For instance:

- If distress is not detected in older adults, the AI may miss them.
- Insufficient representation of minority groups may lead to inaccuracies in the recommendations.
- Predominance of specific language patterns in the data may result in misinterpretation of others.

Bias in mental health instruments can directly impact health outcomes and accessibility to care. The government must address these dangers by mandating assessments of fairness and accuracy.

4.3. Data Proprietorship and Governance

Data governance determines access to data, its permissible uses, and the beneficiaries of its value. In NHS environments, data governance typically adheres to stringent regulations. Nonetheless, the situation becomes more intricate when commercial enterprises develop AI technologies. Certain enterprises store their data on external servers. Some individuals may utilize data to enhance their models or to generate profit. Despite consenting to the terms of service, users may lack a comprehensive understanding of the utilization of their data. The NHS is a public institution that prioritizes the needs of its patients. We require explicit regulations to:

- restrict the utilization of mental health data for ancillary purposes
- ensure transparency in data sharing
- permit users to delete or export their data
- prevent enterprises from exploiting emotional disclosures
- constrain training on user conversations without unequivocal consent

In the absence of such protections, individuals may lose confidence in mental health services, thus hindering their access to assistance.

4.4. Data Transmission and Interoperability

Information circulates among participants in an ecological system. AI tools frequently serve as the initial phase in NHS mental health pathways. Upon completion, the outcomes of an AI-assisted evaluation may be directed to NHS referral systems, physician dashboards, or triage instruments. The AI occasionally provides risk assessments that influence healthcare decisions. Interoperability presents advantages and disadvantages:

- Inadequate integration can alter information.
- Excessive dependence on automated scoring may impair clinical judgment.
- Errors in model output can propagate across the system.
- Sensitive private data may disseminate more broadly than intended.

The ecological perspective demonstrates that AI operates within a broader context. It integrates into a sequence of data interactions that alter tangible entities.

4.4.1. Value Generation and Value Appropriation

AI mental health solutions operate inside a complex value chain connecting developers, NHS commissioners, physicians, and patients. Digital tools facilitate assessments, reduce wait times, and enable earlier interventions, so enhancing the NHS's provision of public services. Simultaneously, private developers derive revenue from user data, enhanced models, and commercial agreements.

Some businesses improve their systems by using data from many users, which makes the product better but also raises moral questions about whether patients really agree to training future models. The NHS creates public value by making it safe and fair for everyone to get digital help. However, this depends on strong governance to make sure that business interests do not come before patient interests. To formulate policies that reconcile innovation with public responsibility, it is essential to understand the flow of value throughout this chain.

4.5. Affective Data and Susceptibility

Information regarding mental health transcends mere technicality. It represents vulnerability, suffering, optimism, and trepidation. AI systems must manage this responsibly. Certain individuals experience a sense of security when conversing with chatbots and develop emotional bonds with them. This emotional bond influences their behaviour and the information they provide. The utilization of AI as a replacement for social connection raises ethical concerns about dependency.

Artificial intelligence is incapable of experiencing empathy. It responds with phrases that were encoded based on established patterns. Should individuals depend on those responses for solace, they may refrain from engaging with others, thereby exacerbating their feelings of isolation. Data ecology acknowledges the mutual impact of technology on behaviour and behaviour on data. An iterative cycle may emerge wherein emotional dependence results in increased online engagement, thus damaging real-life connections.

4.6. Societal Implications

AI mental health tools significantly impact society at large. When utilized appropriately, they can facilitate timely assistance and reduce stigma. They assist individuals who feel marginalized by conventional healthcare or prefer not to engage in face-to-face communication. They may assist individuals in feeling less isolated by directing them towards community support. However, societal dangers include:

- normalizing automated emotional support
- weakening human connections
- exacerbating digital inequality
- reducing privacy standards
- increasing AI's lack of accountability in the event of failure

An increasingly automated counselling society endangers the delegation of mental health support responsibilities from communities and public services. Individuals may perceive a necessity to address their emotional issues independently via digital resources rather than seeking support from others.

5. Conclusion

The NHS employs artificial intelligence to assist individuals with their mental health via an increasing number of AI-driven technologies. This is particularly crucial at present, as an increasing number of individuals are experiencing loneliness and require psychological assistance. These services provide immediate assistance, alleviating the burden on NHS services that are operating at maximum capacity. When utilized appropriately, AI serves as an effective instrument for early intervention by accelerating service delivery and enhancing mental health outcomes. There are potential advantages and significant hazards. AI systems fail to recognize emotional cues, inadequately manage emergencies, and provide users with erroneous or hazardous advice. The systems require extensive private information, which prompts concerns around privacy, data control, and permission. AI techniques inadvertently create inequities due to insufficient representation of minority groups in the training data. Individuals may exhibit diminished motivation in forming genuine friendships if they depend on AI-mediated discussions for communication. This would adversely affect the social well-being that mental health interventions aim to enhance. The identified risks indicate that mental health support requires both professional expertise and robust interpersonal interactions. Technology must acknowledge and honour the inherent boundaries that exist among individuals.

The UK Government bears primary responsibility for this scenario. Promotion and regulation must operate as cohesive systems. The introduction of new technology without adequate monitoring methods can lead to their permanent integration into public services, posing significant risks. If enterprises solely concentrated on limitations, they would be unable to identify other methods to enhance their access systems. We require a measured and meticulously considered strategy. The government ought to allocate funds to evidence-based digital tools, facilitate the advancement of NHS digital infrastructure, establish robust data protection regulations, and ensure the implementation of transparent initiatives. The system must monitor AI operations and establish guidelines for when human specialists should intervene. It must also monitor unforeseen societal repercussions.

AI technology ought to augment human healthcare instead of replacing the entire process. The organization exists to provide assistance and counsel while disseminating knowledge; nevertheless, it must not supplant the vital relationship between patients and their therapists or the essential networks of community support. To enhance mental well-being, individuals must engage with others, demonstrate empathy, and cultivate trust. AI systems facilitate these activities; nevertheless, they do not possess the ability to do these jobs independently. The process of appropriate integration requires the maintenance of human judgment and emotional comprehension as its essential elements.

In conclusion, the UK Government must foster innovation while ensuring robust safeguards are established for the effective utilization of artificial intelligence in NHS mental health services. AI solutions can facilitate early intervention, enhance service accessibility, and alleviate the strain on overburdened services when governed by explicit regulations, transparency, and human supervision. However, if these technologies are not managed appropriately, they may undermine trust, exacerbate inequality, and diminish emotional support in human connections. An equitable policy framework prioritizing safety, fairness, and accountability will enable AI to assist with mental health while preserving the essential human elements of well-being.

6. Hypotheses for Future Research

1. AI-assisted mental health assessments may be more effective in identifying individuals in distress at an early stage when integrated with community-based support.
2. Transparent and accessible data governance could significantly enhance public trust in the NHS's digital mental health solutions.
3. A hybrid approach combining AI and human counselling may prove more effective in the long term than relying only on either AI or traditional approaches.
4. AI systems can alleviate feelings of loneliness by facilitating connections to real-life social activities rather than supplanting human interaction.
5. Algorithms designed to mitigate bias could enhance the accuracy of AI tools for underrepresented or minority groups.
6. Constant monitoring of AI systems may reduce the likelihood of adverse responses during crises.
7. Enhanced initiatives for instructing individuals in technology utilization could mitigate the adverse impacts of digital inequality in AI mental health services.
8. AI systems may alter public expectations on mental health care, thereby influencing their approaches to seeking assistance.
9. Data visualization technologies may assist physicians in comprehending the insights derived from AI.
10. Ethical design guidelines may reduce the likelihood of individuals depending on AI emotional support systems.

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Instructor: Dr. Yang Li

Student Name: Keshav Poudel

Student ID: 3064995

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Part-2: Session

Worksheet 1: Online Resources

This worksheet has a list of useful online resources from the UEL Library and other academic sites that can help you learn and do research in Data Science, statistics, and R.

Category	Title	URL
UEL Subscription Journal	Publications of the American Statistical Association	https://www.tandfonline.com/loi/uasa20
UEL Subscription Journal	Medical Image Analysis (Elsevier)	https://www.sciencedirect.com/journal/medical-image-analysis
UEL Subscription Journal	Operations Research (INFORMS)	https://pubsonline.informs.org/journal/opre
UEL Subscription Journal	Bulletin of the American Society for Information Science and Technology	https://asistdl.onlinelibrary.wiley.com

Table 1: UEL Subscription Journals

Category	Title	URL
Open Access Journal	Harvard Data Science Review	https://hdsr.mitpress.mit.edu
Open Access Journal	EPJ Data Science	https://epjdatascience.springeropen.com
Open Access Journal	IEEE Access	IEEE Xplore: IEEE Access

Open Access Journal	The R Journal	https://journal.r-project.org
Open Access Journal	Bioinformatics (Oxford Open)	https://academic.oup.com/bioinformatics

Table 2: Open Access Journals

Category	Title	URL
Blogging / Resource Site	R-Bloggers	https://www.r-bloggers.com
Blogging / Resource Site	Towards Data Science	Towards Data Science
Blogging / Resource Site	Data Science Central	https://www.datasciencecentral.com
Blogging / Resource Site	KDnuggets	https://www.kdnuggets.com

Table 3: Blogging / Resource Sites

Category	Title	URL
Data Resource	Office for National Statistics (ONS)	https://www.ons.gov.uk
Data Resource	London Datastore	https://data.london.gov.uk
Data Resource	UK Data Service	https://ukdataservice.ac.uk

Table 4: Data Resources

Tutorial / Textbook	Beginning Data Science in R (Thomas Mailund, 2017 / 2022)	https://link.springer.com/book/10.1007/978-1-4842-4354-1
R Tutorial / Textbook	Data Science in Education Using R (Estrellado, Routledge, 2021)	https://www.routledge.com/Data-Science-in-Education-Using-R
R Tutorial / Textbook	R 4 Data Science Quick Reference (Mailund, 2022)	https://www.apress.com/gp/book/9781484272054
R Tutorial / Textbook	Advanced Statistics with Applications in R (Wiley, 2020)	https://www.wiley.com/en-gb/Advanced+Statistics+with+Applications+in+R
R Tutorial / Textbook	Mastering Text Mining with R (Packt, 2016)	https://www.packtpub.com/product/mastering-text-mining-with-r/9781784397005

Table 5: R Tutorial / Textbook Resources

Worksheet 2: Tech Clusters mini-project

A Comparison of UK Technology Clusters: London and Cambridge in Terms of Data Ecology

Introduction

This mini-project uses a data ecology point of view to critically compare the technology clusters in London and Cambridge. The analysis looks at how data is made, shared, controlled, and used in each ecosystem, rather than just looking at clusters as groups of businesses. The comparison shows how differences in the amount of data, the way institutions are set up, and the way they are run affect the UK's information economy's ability to innovate, create economic value, and stay in business for a long time.

The environment for making data and information

The size and variety of data in London's technology cluster set it apart. Financial services, transportation, healthcare, and digital platforms are just a few of the sectors that constantly create large amounts of transactional, behavioural, and administrative data. Government-led open-data projects, like those that provide transportation and demographic data, help new businesses get started by making it easier for them to try new things. For instance, fintech companies in London depend a lot on real-time financial and consumer data, which is made useful through aggregation and analytics instead of new scientific ideas. The data environment at Cambridge is more focused on research and less on general use. The University of Cambridge and its affiliated research institutes, especially those that study life sciences, biotechnology, and artificial intelligence, are very important to data production. Here, the value of data comes from how accurate, original, and scientifically credible it is, not how much of it there is. For example, biomedical spin-outs often work with very tightly controlled experimental and clinical datasets, where protecting intellectual property and governance are very important to the data ecology.

Individuals, Regulations, and Framework

The data ecosystem in London comprises various stakeholders, including government agencies, regulators, multinational technology firms, start-ups, investors, and a substantial digital workforce. This fosters intense competition, and data circulates rapidly across firms. This transparency complicates governance due to concerns regarding privacy, adherence to regulations, and the power disparity between large platforms and small enterprises. Regulatory authorities significantly influence data utilization, particularly in sectors such as healthcare and finance.

The ecology in Cambridge is more interconnected and institutionally grounded. Universities, research laboratories, and spin-off enterprises are the primary generators and consumers of data. Governance systems are more defined and regulated, which diminishes the dangers of legal violations or unethical conduct; nevertheless, it also complicates data sharing across sectors. Infrastructure prioritizes depth over width, indicating that enduring scientific partnership is preferable to rapid commercial expansion.

Potentialities and Constraints

The primary advantage of London lies in its network effects. Access to funding, skilled personnel, and diverse datasets facilitates rapid innovation and growth. However, the factors that facilitate ease also introduce challenges, such as elevated operational expenses, environmental strain on digital infrastructure, and increasing concerns around data ethics and monitoring. Smaller enterprises may struggle to compete with larger platforms that possess extensive data resources.

Cambridge possesses substantial trust, effective information transmission, and consistent funding for research. Its primary issue is its limited capacity for growth. Housing shortages, transportation challenges, and a diminished labour pool impede corporate growth. Furthermore, dependence on specialized data complicates the ability of organizations to experiment across several areas.

Conclusion

From a data ecology perspective, London and Cambridge represent two distinct evolutionary models within the UK technology ecosystem. London is a vast, data-intensive ecosystem characterized by its scale, diversity, and rivalry. Nevertheless, it has significant challenges regarding governance and sustainability. Cambridge is a specialized, knowledge-driven ecosystem that prioritizes data quality, governance, and sustained innovation over scale. They demonstrate that successful technological clusters rely on factors beyond their geographical location. Their effectiveness is contingent upon the configuration, operation, and alignment of data ecosystems with social and economic objectives.

Worksheet 3: Data Integration

Goal

The aim of this initiative was to compile and combine open administrative and census data for analytical purposes. Individual property transaction records from HM Land Registry were integrated with aggregated census dwelling stock data to analyse regional trends in housing market activity. The emphasis was on data cleansing, utilization of lookup tables, aggregation using pivot tables, and computation of comparable proportions across areas.

Data preparation and integration

Property transaction data for England and Wales were obtained from HM Land Registry and refined to include only finalized transactions from the most recent complete month and valid records. District names were standardized to facilitate alignment with official local authority designations. A lookup table for local authority to region was utilized to designate regional geography. The Census 2021 dwelling stock data (TS044) were organized to correspond with transaction property types and aggregated by area. Both datasets were consolidated using pivot tables and merged into a singular regional table.

Districts	Properties sold				Base dwelling stock				Percentage sold			
	Sum of D	Sum of F	Sum of S	Sum of T	Sum of detached	Sum of semi-detached	Sum of terraced	Sum of flats/maisonettes	%_D	%_F	%_S	%_T
East	1436	872	1576	1415	776813	816068	557952	381096	0.002	0.002	0.002	0.003
East Midlands	1197	540	1224	1135	676022	727739	392643	180577	0.002	0.003	0.002	0.003
London	907	1385	1210	1424	208078	636077	727671	1380140	0.004	0.001	0.002	0.002
North East	438	276	528	418	207944	470848	326453	133228	0.002	0.002	0.001	0.001
North West	1294	654	1695	1438	599838	1160270	886330	394014	0.002	0.002	0.001	0.002
South East	2392	1935	2547	2468	1065035	1082661	812587	640091	0.002	0.003	0.002	0.003
South West	1090	1104	1049	1057	734776	694153	546602	314382	0.001	0.004	0.002	0.002
Wales	573	408	699	929	384397	431871	358034	126663	0.001	0.003	0.002	0.003
West Midlands	12675	9738	14097	13602	595851	913117	523411	318415	0.021	0.031	0.015	0.026
Yorkshire and The Humber	644	777	630	766	504506	877371	607775	257823	0.001	0.003	0.001	0.001
England & Wales	22646	17689	25255	24652	5753260	7810175	5739458	4126429	0.004	0.004	0.003	0.004

Table 6: Regional proportions of property transactions by dwelling type

Interpretation

The findings indicate that the West Midlands possesses the highest ratio of property transactions to its housing supply across all property categories, signifying a more dynamic real estate market in comparison to other regions. The South West and South East have comparatively elevated proportions of specific home forms, notably terraced houses and flats/maisonettes. Conversely, London exhibits a lower relative turnover despite substantial absolute transaction volumes, indicative of its extensive housing stock. Regional disparities in transaction proportions underscore differing property market dynamics throughout England and Wales.

Worksheet 4: Google Trends

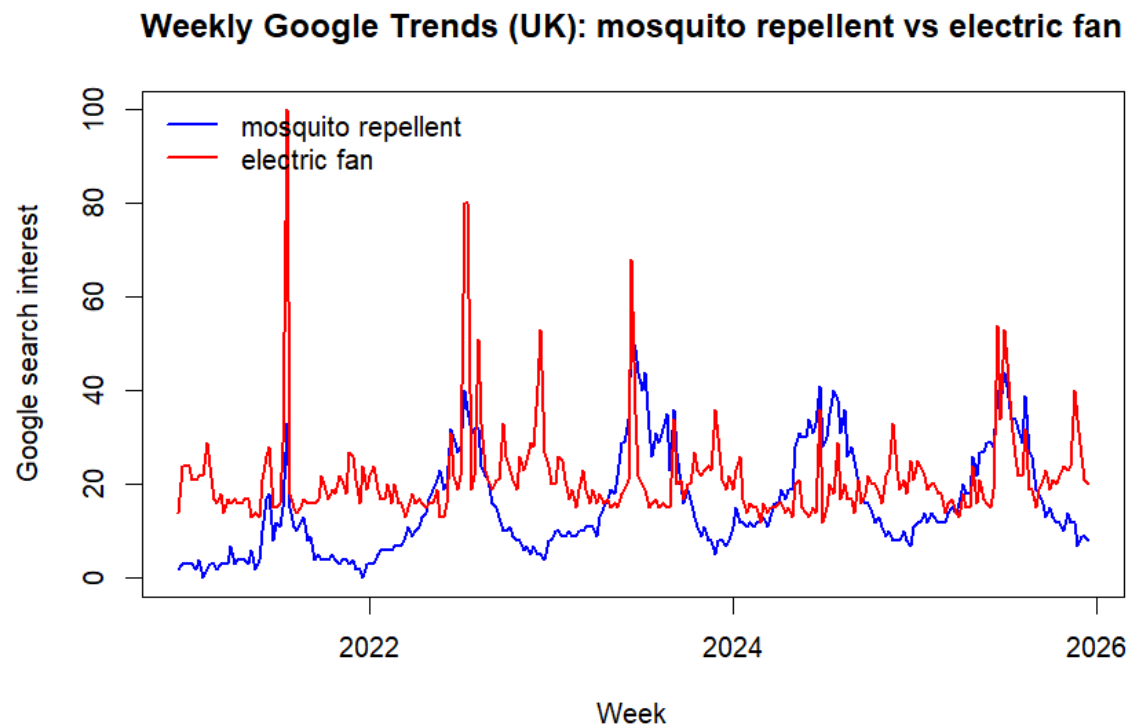


Figure 1: Google trends Mosquito vs Fan UK Weekly

Correlation result (from R):

- Pearson $r = 0.300$
- $p < 0.001$
- 95% CI: [0.19, 0.41]

Interpretation

The weekly Google Trends data for mosquito repellent and electric fan exhibit a positive albeit lesser association ($r = 0.300$, $p < 0.001$) in comparison to the monthly data. Despite both searches increasing during warmer seasons, short-term changes and periodic weekly variations diminish the strength of the correlation. This indicates that the fundamental seasonal signal is more distinct at reduced temporal resolution.

Monthly Google Trends (UK): mosquito repellent vs electric fan

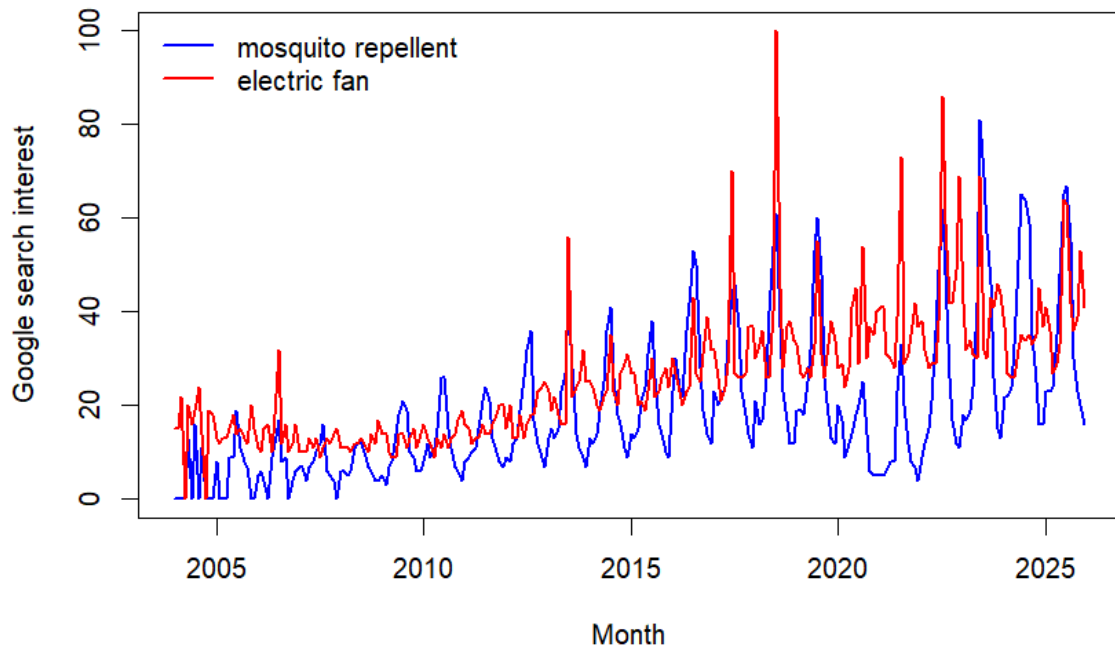


Figure 2: Google trends for Mosquito vs Fan UK Monthly

Correlation result (from R)

- **Pearson $r = 0.569$**
- **$p < 0.001$**
- **95% CI: [0.48, 0.65]**

Interpretation

The monthly Google Trends data for mosquito repellent and electric fan exhibit a moderate to strong positive association ($r = 0.569$, $p < 0.001$). Both searches grow during warmer months and diminish in winter, indicating a genuine correlation influenced by seasonal temperature and discomfort. This association is significant as both variables react to the same real-world environmental influence.

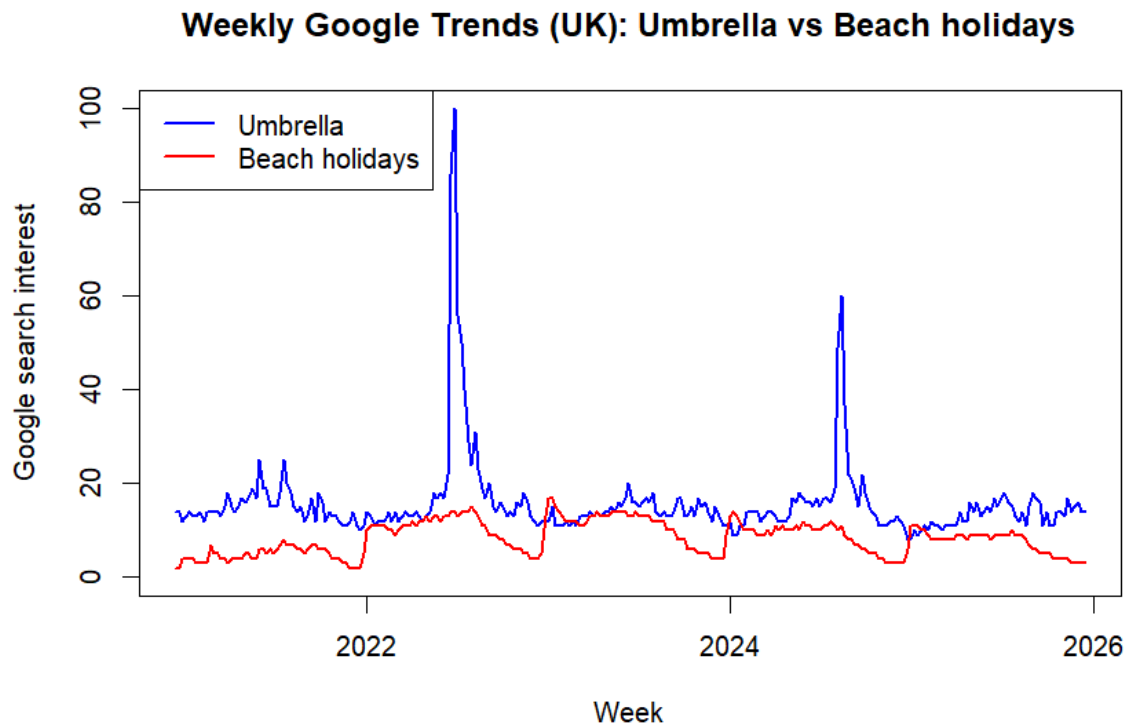


Figure 3: Google trends for umbrella and beach holiday weekly

Correlation result (from R)

- **Pearson $r = 0.42$**
- **$p < 0.001$**
- **95% CI: [0.32, 0.51]**

Interpretation

The weekly Google Trends data for umbrellas and beach holidays exhibit a moderate positive connection ($r = 0.42$, $p < 0.001$). In comparison to monthly data, the correlation is diminished owing to short-term fluctuations and interference in weekly searches. Spikes frequently align with particular meteorological occurrences rather than enduring seasonal trends. Nonetheless, the favourable correlation suggests that both search phrases react to identical short-term fluctuations in precipitation.

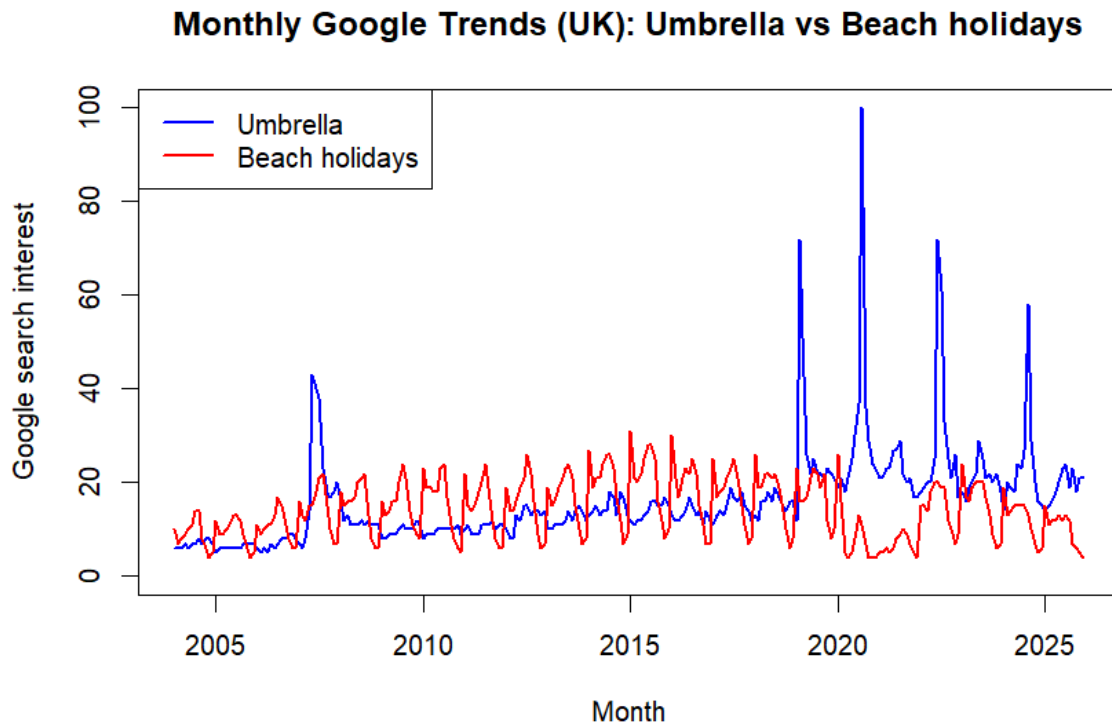


Figure 4: Google trends for umbrella and beach holiday monthly

Correlation result (from R)

- **Pearson $r = 0.61$**
- **$p < 0.001$**
- **95% CI: [0.55, 0.67]**

Interpretation

The monthly Google Trends data for umbrellas and beach holidays exhibit a significant positive association ($r = 0.61$, $p < 0.001$). Interest in both phrases increases during wetter seasons and diminishes during dry months. This suggests that public search behavior for rain-related content is consistently aligned over extended time periods. The correlation's intensity indicates a common environmental factor, specifically rainfall, rather than a clear causal link between the two search phrases.